

Assessment of Groundwater Quality Using Water Quality Index (WQI), Pollution Index of Groundwater (PIG), and GIS in Parts of Anantapur District, Andhra Pradesh, India

P. Ravikumar¹, S. Srinivasa Gowd^{2*}, C. Krupavathi³, G. Harish Vijay⁴

Abstract

The purpose of this study was to assess the potability of groundwater resources in some parts of the Anantapur district of the Indian state of Andhra Pradesh. The water quality index, a spatial evaluation of groundwater using a GIS-based interpolation method, and the polluting groundwater index are some of the methods used to analyze water quality and explain the water quality of the studied area. A total of 22 groundwater samples were obtained during the post-monsoon period. The samples were tested for pH, EC, TDS, TH, and the major cations and anions. PIG levels between 0.98 and 1.98, with an average of 1.36, categorize low contamination in 5% of the research area and groundwater that is fit for human consumption. Additionally, 77% of the groundwater samples have drinking qualities that are only slightly acceptable. WQI values ranged from 85.97 to 113.52 mg/L, indicating that 55% and 45% of the samples are safe for drinking, respectively, while the other 45% are not. Groundwater samples are found in the field of rock dominance, according to the Gibbs plot. GIS techniques are being used to analyze the spatial variation of groundwater quality, and the results show that the majority of groundwater samples marginally meet standards for potable water, necessitating prior treatment before use.

Keywords: Groundwater, Hydro geochemistry, PIG, WQI, GIS, Spatial distribution Maps.

INTRODUCTION

Water is necessary for human survival, as well as all forms of development. Huge population expansion, increased urbanization, expanding industries, and massive agricultural operations around the world have all contributed to a massive increase in the mandate for freshwater for home, agricultural, and industrial uses in recent years [1–2]. Groundwater is a life elixir, but it has been plagued by challenges in recent years, particularly in terms of quality, particularly in dry and semiarid areas. Agricultural pollutants, such as chemicals, insecticides, and other metal contamination, make water unsafe for human consumption. As a result, understanding the hydrogeochemistry of water is

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critical for evaluating water for drinking, agriculture, and other uses. Understanding groundwater occurrence, replenishment, and recovery is especially critical in dry and semi-arid areas because of disparities in monsoonal precipitation, lacking shallow waters, and over-drafting of groundwater resources. Aquifer quality is affected by recharged water, meteorological rainfall, inland groundwater, and underground geochemical processes. As a result, knowing hydrochemistry is crucial for evaluating groundwater quality in any location where groundwater is used for drinking [3–4]. Water excellence testing can reveal important evidence

about the subsurface geologic conditions in which the water is found. Groundwater quality has been greatly impacted by over-exploitation. A unified approach of PGI, WQI and the GIS can be used to give a simple and valuable tool for decision-making on groundwater excellence [5–6]. The pollution index of groundwater (PIG) is a useful tool for assessing the appropriateness of potable water in any region and disseminating data on general quality of water [7–8]. WQI is a scientific tool that can convert a significant quantity of data on water superiority into a solitary amount that signifies the level of water excellence. Several investigators have previously used the WQI as a tool to obtain the excellence of groundwater [9–10]. GIS can be used to map the spatial distribution of various hydrochemical parameters. It has already proven to be effective for a number of authors [11–13]. The main objective of the study is to evaluate the fitness of water by using the Water Quality Index (WQI), Pollution Groundwater Index, and Geographic Information System (GIS) applications to comprehend the quality distribution. The current paper also discusses the groundwater quality in some areas of Anantapur, Andhra Pradesh.

Study region

The study area is situated center portion of Anantapur district, Andhra Pradesh India, it is enclosed 269 sq. km. This study area geographically lies between $14^{\circ} 25' 0''$ and $14^{\circ} 40' 0''$ N latitude and $77^{\circ} 30' 0''$ and $77^{\circ} 40' 0''$ E longitudes. It has an average elevation of 516 meters (1693 ft). Temperatures in the region range from 24°C to 46°C , and there is an average annual rainfall of 544 mm. The study area falls in the Survey of India Toposheet 57F/6, F/10 and F/11. The study area geologically comprises Hornblende biotite gneiss, hornblende gneiss, biotite gneiss, meta basalts, and grey/pink granites, are the major rocks are found in the Study area. Sample location map of the study area Figure 1.

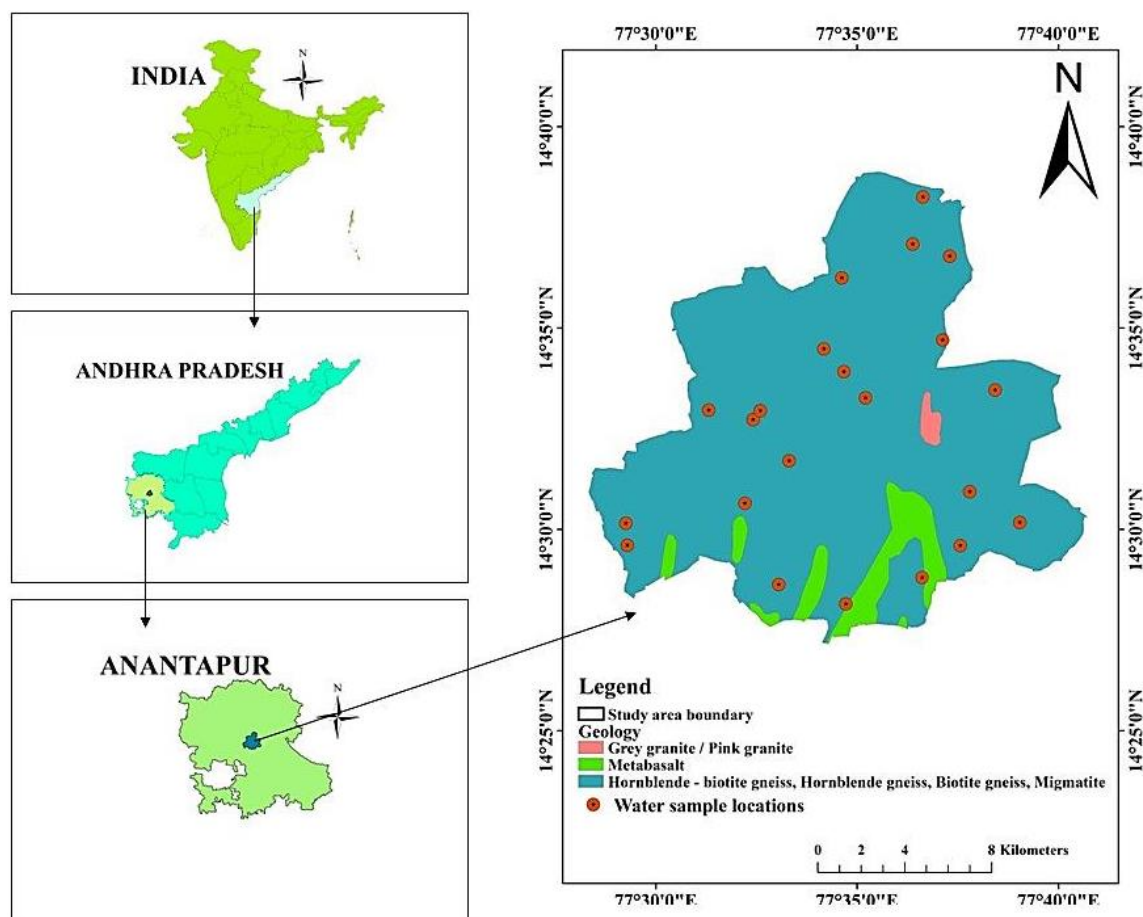


Figure 1. Sample location map of the study area.

MATERIALS AND METHODS

Twenty-two groundwater samples from bore wells and dug wells in the research area were collected to assess the water's quality. The American Public Health Association advised examining groundwater samples for a number of different chemical characteristics [14]. Significant cations including Ca^{2+} , Mg^{2+} , Na^+ , and K^+ as well as anions like, HCO_3^- , Cl^- , and NO_3^- , SO_4^{2-} are present together with pH, EC, TH, TA, and TDS. In the field, pH and EC were determined with a portable conductivity meter. A flame photometer was used to test the levels of Na^+ and K^+ . We measured the NO_3^- and SO_4^{2-} concentrations using a spectrophotometer. TH and TA were examined as CaCO_3 , Ca^{2+} , CO_3^{2-} , HCO_3^- , and Cl^- using the volumetric technique. Using the amounts of TH and Ca^{2+} , Mg^{2+} was calculated. Based on the physico-chemical investigations, certain significant parameters and indices were developed.

Analytical Accuracy

Ionic equilibrium error (IBE) was assessed by comparing the total cation level TZ (Na^+ , K^+ , Ca^{2+} , Mg^{2+}) and total anion level TZ (HCO_3^- , Cl^- , SO_4^{2-} , NO_3^-) for each sample, both expressed in meq/L.

$$IBE = \frac{\Sigma \text{Cations} - \text{Anions}}{\Sigma \text{Cations} + \text{Anions}} \times 100 \text{ (meq/L)} \quad (1)$$

The computed IBE value falls within the allowed range of ± 5 .

Pollution Index of Groundwater

Subba Rao first introduced the pollution index of groundwater (PIG) in the beginning of 2012, and it is now extensively applied to investigate changes in groundwater quality caused by a variety of geochemical variables [15]. The PIG is computed in five steps. The first step is to give each chemical property a relative weight (RW). RW often varies from 1 to 5, based on how much of an influence it has on people's health. Cl^- , SO_4^{2-} , and NO_3^- received the highest weight (5), and K^+ and HCO_3^- received the lowest weight (1). In the second stage, the weight parameter (WP) is generated for each chemical parameter to assess its relative influence to the groundwater's entire purity. The WP is calculated using the formula below:

$$Wp = \frac{Rw}{\Sigma Rw} \quad (2)$$

$$Sc = \frac{c}{Ds} \quad (3)$$

$$Ow = Wp * Sc \quad (4)$$

$$PIG = \Sigma Ow \quad (5)$$

Each groundwater sample's proportionate chemical contribution will be considered for evaluating PIG. Water with an overall chemical quality (Ow) greater than 0.1 equates to 10% of the PIG's 1.0-point value. This is a straightforward explanation of the impact of groundwater body pollution [15]. According to Table 2, the PIG has been categorized as an insignificant contamination (<1.0), low contamination (1.0 - 1.5), moderate contamination (1.5–2.0), high contamination (2.0–2.5), and very high contamination (>2.5). Additional chemical variable intensities are used to describe the gradual infiltration of airborne particles into an aquifer system.

Water Quality Index

WQI computations are divided into 3 stages. The first stage is to "assign weight" to all of the 13 criteria by assigning a weight (wi) depending on how important they are in relation to groundwater resources [16–17]. The second stage involves calculating relative weights using the following equation.

$$Wi = \frac{wi}{\Sigma_{i=1}^n (wi)} \quad (6)$$

The "quality rating (qi)" for the third stage is calculated using the calculation below.

$$qi = \left(\frac{ci}{si} \right) \times 100 \quad (7)$$

If Ci represents the concentration of each characteristic in each water sample, then Si represents the recommended WHO 2012 value for each characteristic [18–19]. The subsequent equations will calculate WQI because Wi and qi were combined to estimate the SII for each characteristic independently.

$$SII = wi \times qi \quad (8)$$

$$WQI = \sum_{i=1}^n SII \quad (9)$$

The sub index of each parameter is here designated as SII.

Spatial Analysis

The spatial analyst and geostatistical analyst modules of Arc GIS 10.8 can be used to compute the spatial changes of groundwater quality based on GIS after the geo database has been created. The weighted inverse distance interpolation approach was used to achieve this (IDW) [20–21]. IDW is a spatially interpolating value technique that can be computed between observations. Every value is the weighted average of the points nearby. From the location observation to the estimation point, weights are computed using the inverse distance approach. The mapping and analysis software tool is used to interpolate the IDW method, which is a useful tool for developing spatial variation maps of these region's subsurface physico-chemical characteristics [22–24]. The IDW approach has been widely employed throughout the world for numerous reasons, creating spatial distribution charts that considerably discriminate intensity regions of high, medium, and low chemical components.

RESULTS AND DISCUSSION

The pH of the groundwater in the research region varied from 7.2 to 8.3, with an average of 7.9, implying that it is moderately alkaline in nature (Table 1). Nearly all samples of groundwater taken at different pH levels fall within BIS's permitted range (2012). Figure 2 illustrated spatial variation of pH. The range of the EC concentration is 890 to 2580 S/cm, with a mean value of 1616.6 μ S/cm. Figure 3 shown spatial distribution of EC. TDS up to 500 mg/L is safe [25]. The bulk of the groundwater samples used in the investigation were inappropriate for household use because 100% of the entire sample had TDS concentrations that were higher than the BIS (2012) permissible level for drinking water. Spatial variation map of TDS (Figure 4). The study's average groundwater TH ranges from 349 to 936 mg/L, surpassing the recommended drinking water limit of 200 mg/L [25], with the majority of the groundwater samples having THs above this value (Table 1) Figure 5 depicts the spatial variation of TH.

Cation Anion Chemistry

The most common cations are Na^+ and Ca^{2+} , followed by Mg^{2+} and K^+ , according to the cation chemistry. Na^+ content in this study differs from 92 to 218 mg/L, with a mean value of 137 mg/L (Table 1 and Table 2). The upper permissible amount is 200 mg/L, as per WHO (2012) recommendations [26]. Only two sampling locations are higher than the permissible level for drinking purposes, according to this study (Table 2). The Ca^{2+} levels range between 84 to 188 mg/L (Table 1). The average Ca^{2+} concentration is 125.9 mg/L, exceeding the permissible limit of 75 mg/L recommended for potable water in all groundwater samples. The ranges of Mg^{2+} and K^+ concentration are 18–122 mg/L and 3–16 mg/L, respectively, with mean values of 60.9 and 7.5 mg/L (Table 1 and Table 2). Furthermore, the total amount of Mg^{2+} in the groundwater samples is 90% greater than the permitted level of 30 mg/L. HCO_3^- is the most abundant anion in the research area, followed by Cl^- , SO_4^{2-} , and NO_3^- . The Cl^- level of groundwater ranges from 122 to 249 mg/L, with a mean of 180.5 mg/L. All groundwater samples fall below the 250 mg/L potable water standard value BIS 2012.

HCO₃⁻ levels ranged from 389 to 732 mg/L, however there is no recommended standard value for HCO₃⁻. The average groundwater concentration of SO₄²⁻ in the research area was 117.9 mg/L, and 95% of the samples were below the 200 mg/L drinking-water safe-level. The content of NO₃⁻ in the groundwater used for this study ranges from 15 to 45 mg/L, with a mean of 27.9 mg/L; its higher content (>10 mg/L) suggests the effects of waste water discharge and nitrogen fertilizer use on the aquifer system [27]. Figures 6, 7, 8, 9, 10, 11, 12, 13 illustrate the spatial distribution of all anions (HCO₃⁻, Cl⁻, SO₄²⁻, NO₃⁻) and cations (Ca²⁺, Na⁺, Mg²⁺, K⁺).

Table 1. Analytical results of physico-chemical parameters of the study region.

S.N.	Sample locations	pH	EC	TDS	Ca ²⁺	Mg ²⁺	Na ⁺	K ⁺	HCO ₃ ⁻	Cl ⁻	SO ₄ ²⁻	NO ₃ ⁻	TH
1	Bandameedapalle	7.8	1315	680	121	112	104	8	732	160	105	15	762
2	Bommaparthi	8.1	1460	790	188	92	218	5	649	249	156	22	936
3	Bukkacherla	7.9	1020	640	96	84	118	9	488	193	112	24	584
4	Gandlaparthi	8.1	1150	590	85	74	108	3	427	195	97	35	516
5	Gangalakunta	7.5	1530	780	109	82	128	10	489	155	250	40	609
6	Gondireddypalle	8	1750	890	98	45	146	7	468	148	97	32	430
7	Hampapuram	7.8	1680	860	105	48	121	16	621	169	82	22	459
8	Maroor	7.4	890	480	116	32	142	7	549	178	96	28	421
9	Palacherla	8.3	1860	980	182	54	106	6	519	244	90	38	676
10	Rapthadu	7.6	960	510	96	46	118	9	389	172	88	18	429
11	Erragunta	8.1	1380	740	84	74	96	7	448	122	172	42	513
12	Parasannayapalli	7.2	1670	850	122	108	128	9	618	195	85	45	748
13	Gollapalli	8	2360	1220	83	18	198	4	546	142	59	36	281
14	Cherlopalli	7.8	2450	1320	114	28	212	6	524	142	152	32	400
15	Boginepalli	8.2	910	520	121	64	92	4	438	124	137	28	565
16	Pullalarevu	8.2	1660	880	142	38	108	10	448	203	87	32	511
17	Varimadugu	7.5	1840	940	162	62	128	5	554	213	119	28	659
18	G. Kothapalli	7.7	1380	720	124	74	191	12	569	216	185	34	613
19	Ramanepalli	8	2580	1320	174	36	128	10	568	146	113	18	583
20	Palabavi	7.8	2170	1110	168	48	128	8	549	213	93	24	617
21	Ayyavaripalle	7.9	1880	980	124	42	184	5	732	231	78	28	482
22	Chapatla	7.8	1670	880	128	78	112	6	528	160	140	32	640

Table 2. Concentrations of ions and their comparison with the WHO and BIS.

Parameter	Minimum	Maximum	Mean	SD	WHO (2012)	BIS (2012)
pH	7.2	8.3	7.9	0.28	6.5-8.5	6.5-8.5
EC	890	2580	1616.6	485.82	–	–
TDS	480	1320	849.1	242.13	500	500
Ca ²⁺	84	188	125.9	30.81	75	75
Mg ²⁺	18	112	60.9	25.55	30	30
Na ⁺	92	218	137.0	38.16	200	–
K ⁺	3	16	7.5	3.00	–	–
HCO ₃ ⁻	389	732	538.8	91.15	–	–
Cl ⁻	122	249	180.5	37.33	250	250
SO ₄ ²⁻	59	250	117.9	43.97	250	200
NO ₃ ⁻	15	45	29.7	8.03	45	45
TH	281	936	565.2	144.06	–	200

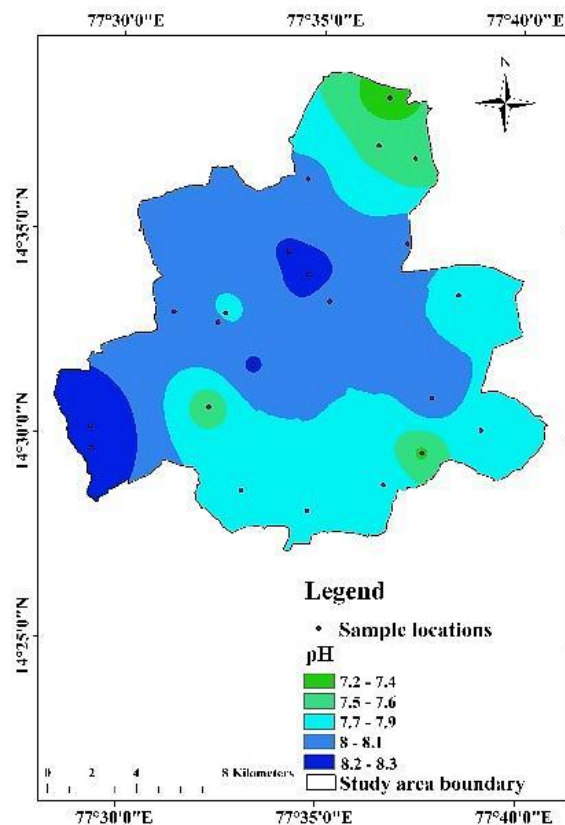


Figure 2. pH

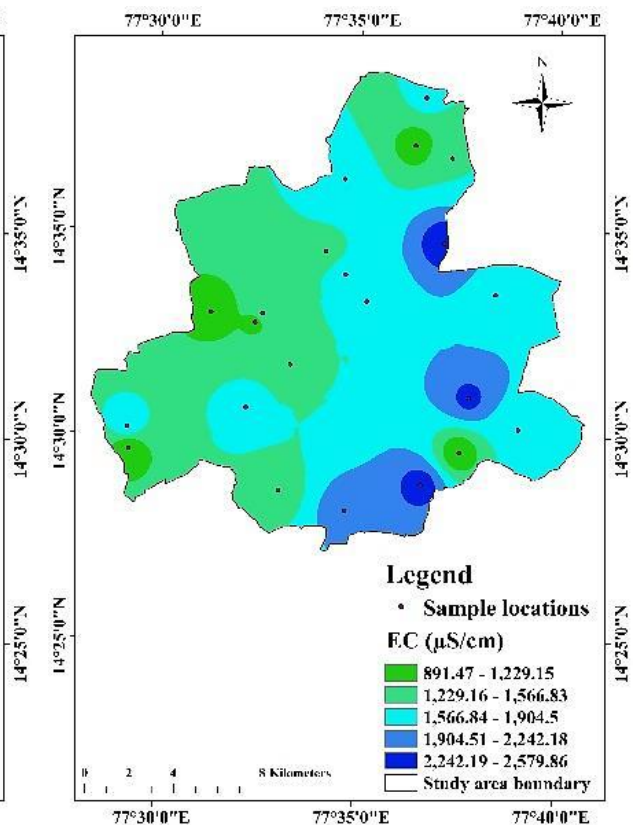


Figure 3. EC

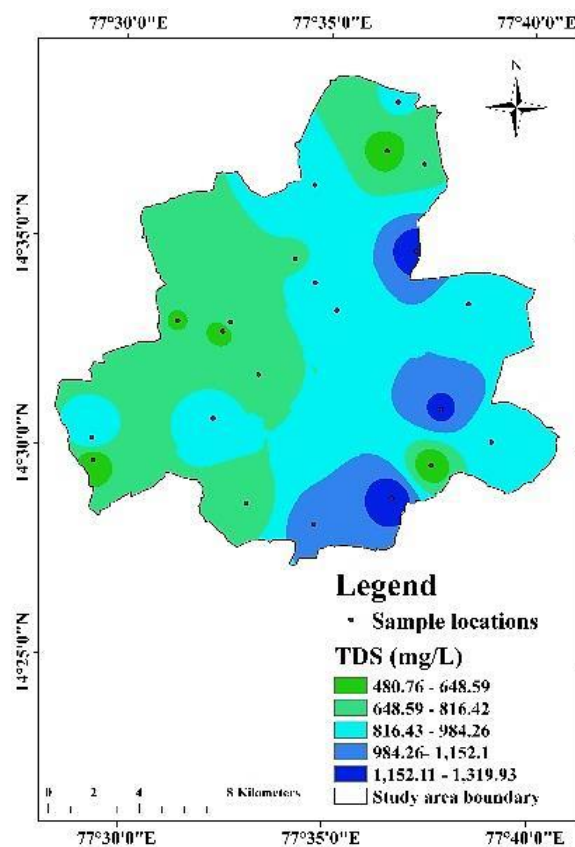


Figure 4. TDS

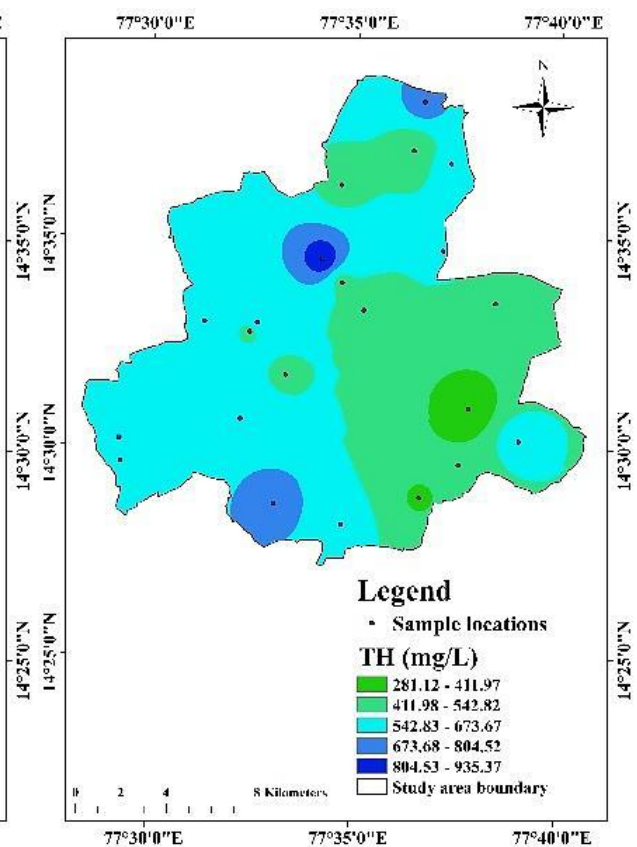


Figure 5. TH

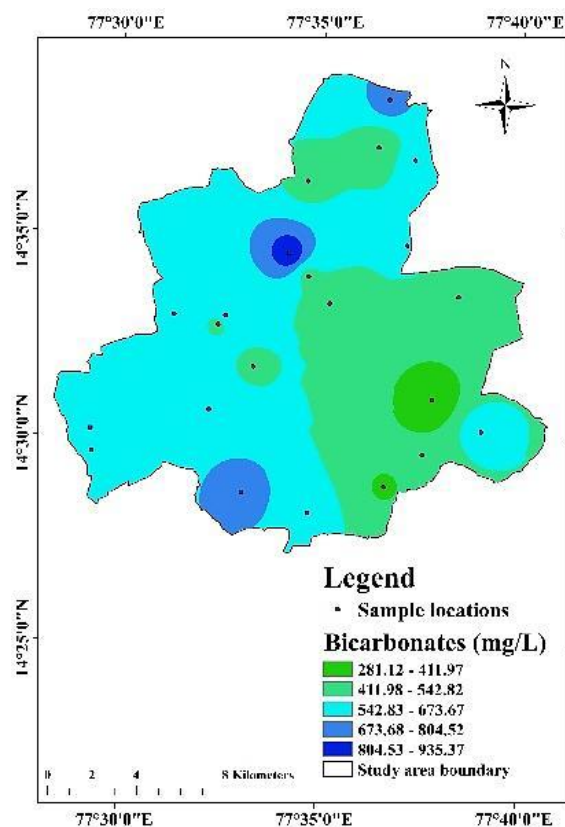


Figure 6. HCO_3^-

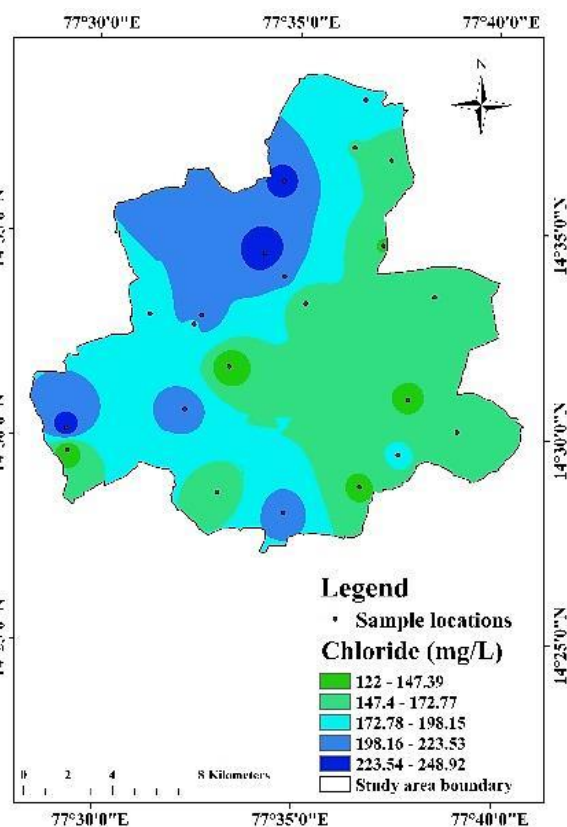


Figure 7. Cl^-

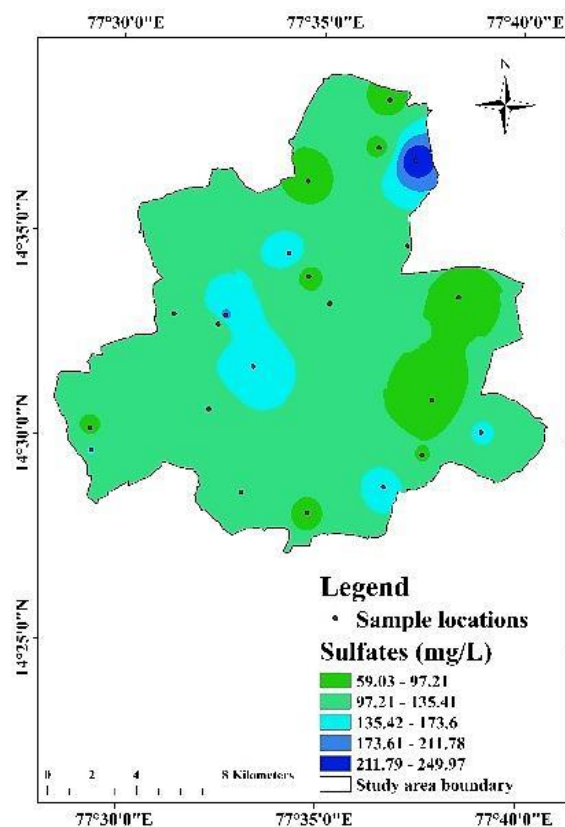


Figure 8. SO_4^{2-}

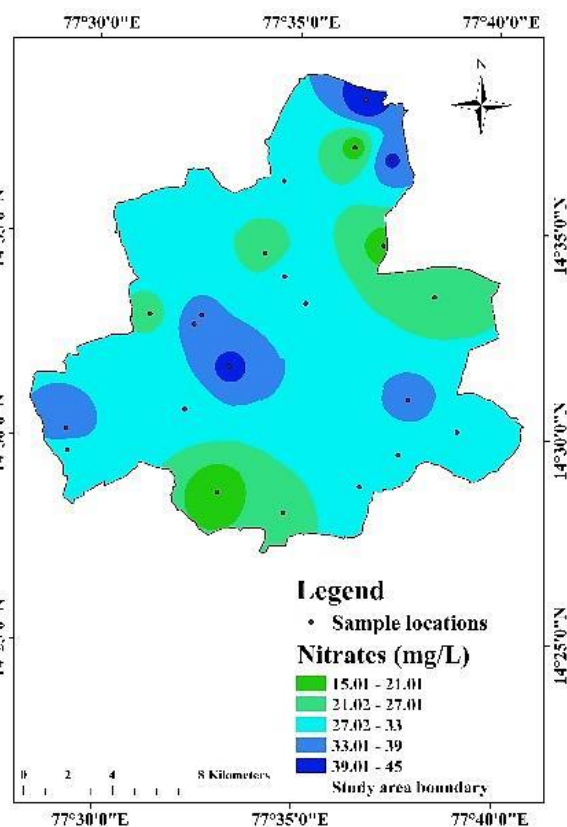
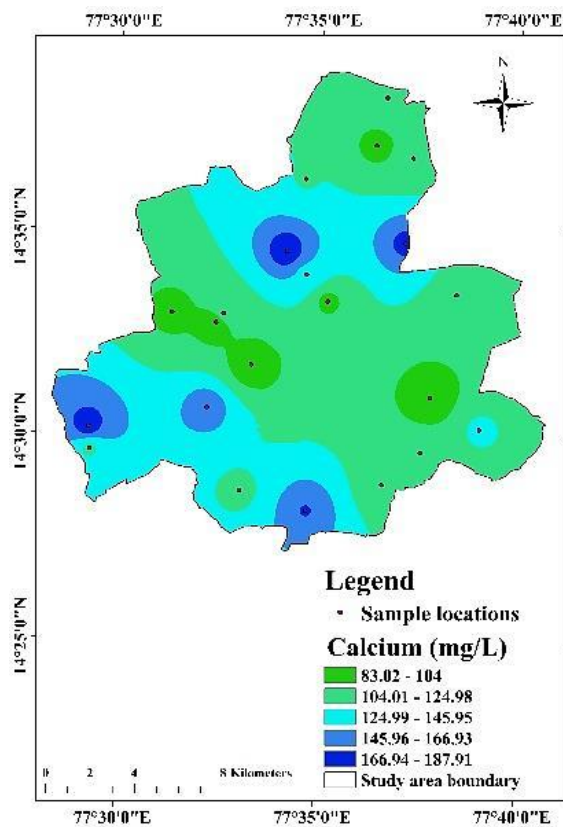
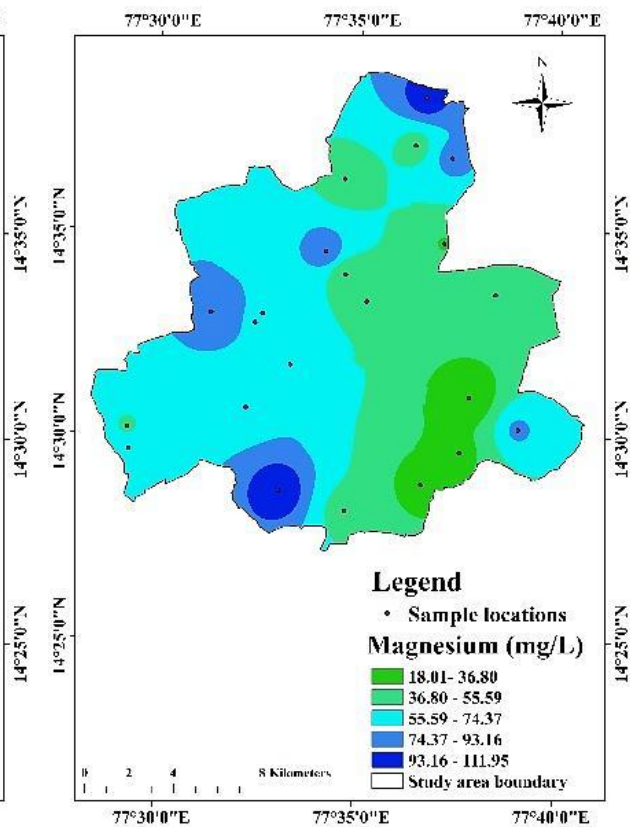
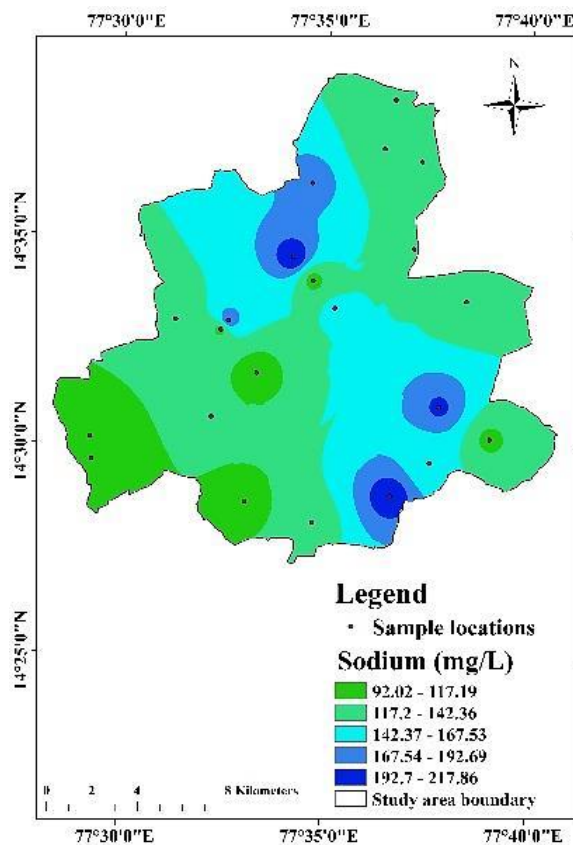
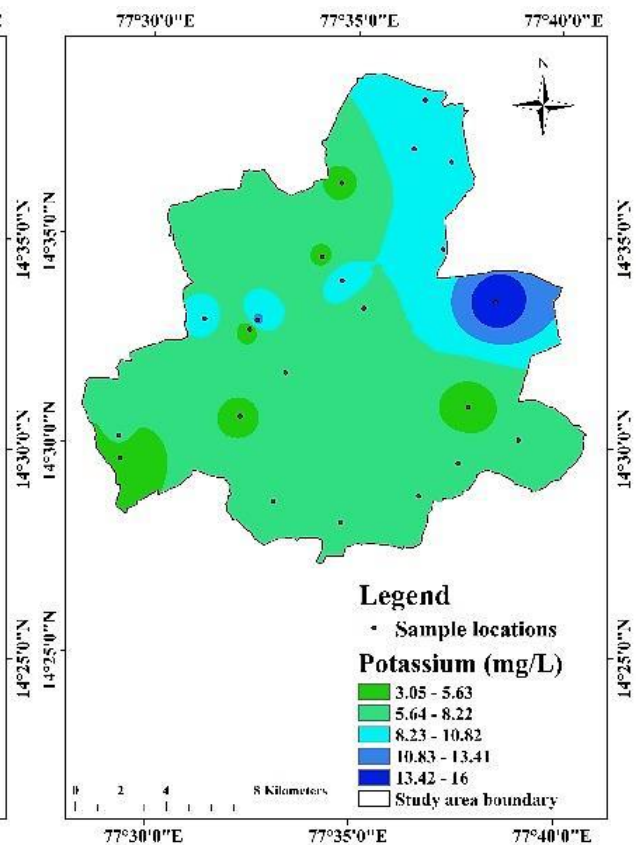


Figure 9. NO_3^-

Figure 10. Ca^{2+} Figure 11. Mg^{2+} Figure 12. Na^{+} Figure 13. K^{+}

Water quality index (WQI)

The Water Quality Index is divided into five categories: excellent type (0–50), good type (50–100), poor type (100–200), extremely poor type (200–300), and unfit for consumption (>300). (Table 3). The range of the WQI is 85.97 to 113.52. Table 4 shows that 55% of the samples have high water quality, whereas 45% have low water quality. Figure 14 shown the spatial distribution map of WQI.

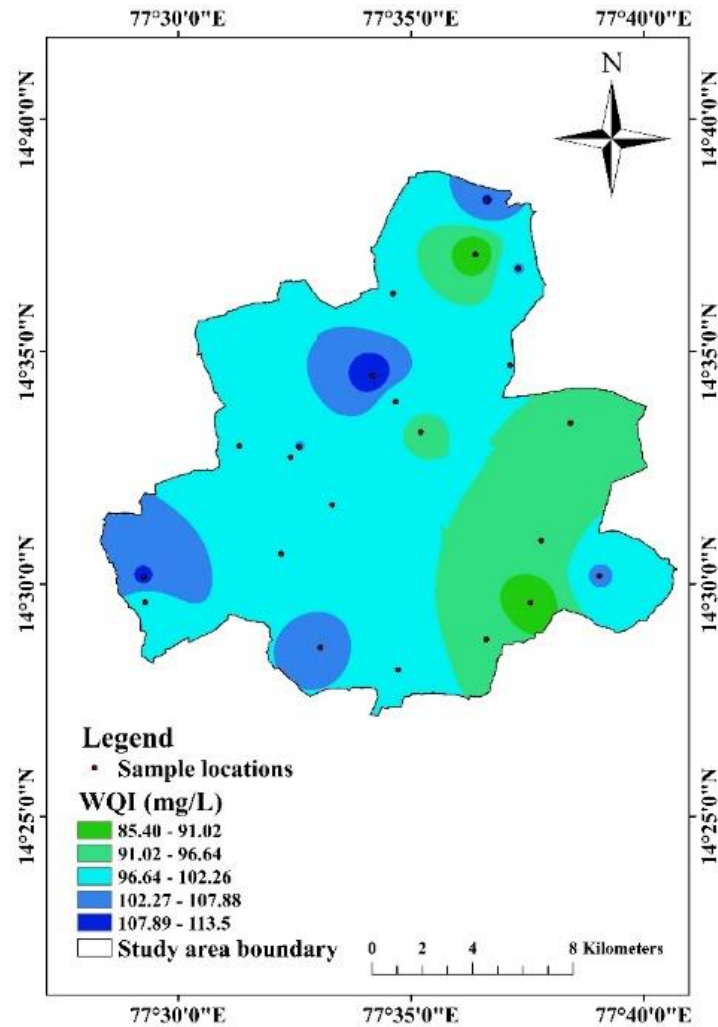


Figure 14. Spatial distribution map of WQI.

Table 3. WQI individual sampling stations.

Sample no	Name of the village	Latitude Value	Longitude value	WQI	Water quality status
1	Bandameedapalle	N 14° 26' 21"	E 77° 38' 3"	104.26	Poor Water
2	Bommaparthy	N 14° 34' 30"	E 77° 34' 28"	113.52	Poor Water
3	Bukkacherla	N 14° 26' 46"	E 77° 31' 33"	98.15	Good Water
4	Gandlaparthy	N 14° 32' 39"	E 77° 31' 3"	98.97	Good Water
5	Gangulakunta	N 14° 29' 24"	E 77° 32' 39"	102.5	Poor Water
6	Gondireddipalle	N 14° 26' 1"	E 77° 29' 8"	94.58	Good Water
7	Hampapuram	N 14° 24' 26"	E 77° 30' 48"	91.77	Good Water
8	Maroor	N 14° 25' 35"	E 77° 37' 28"	85.97	Good Water
9	Palacherla	N 14° 23' 12"	E 77° 30' 4"	109.03	Poor Water
10	Rapthadu	N 14° 36' 56.9"	E 77° 36' 22.8"	85.38	Good Water
11	Yerragunta	N 14° 24' 3"	E 77° 25' 30"	101.34	Poor Water

Sample no	Name of the village	Latitude Value	Longitude value	WQI	Water quality status
12	Parasannayapalli	N14° 25' 16"	E 77° 25' 58"	108.13	Poor Water
13	Gollapalli	N 14° 26' 57"	E 77° 26' 57"	93.37	Good Water
14	Cherlopalli	N 14° 31' 57"	E 77° 26' 56"	94.75	Good Water
15	Boginepalli	N 14° 33' 24"	E 77° 27' 55"	98.17	Good Water
16	Pullalarevu	N 14° 29' 24"	E 77° 27' 56"	98.39	Good Water
17	Varimadugu	N 14° 29' 18"	E 77° 27' 32"	100.78	Poor Water
18	G. Kothapalli	N14° 32' 58.4"	E77° 32' 31.2"	102.64	Poor Water
19	Ramanepalli	N 14° 27' 3"	E 77° 33' 30"	99.12	Good Water
20	Palabavi	N 14° 27' 25"	E 77° 26' 35"	100.29	Poor Water
21	Ayyavaripalle	N 14° 25' 43"	E 77° 26' 30"	96.73	Good Water
22	Chapatla	N 14° 24' 18"	E 77° 29' 81"	103.08	Poor Water

Table 4. Water quality categorization based on WQI value.

Water quality index value (WQI)	Class	Water quality status	Percentage of water samples
<50	I	Excellent water	–
50–100	II	Good water	55%
100–200	III	Poor water	45%
200–300	IV	very poor water	–
>300	V	Unsuitable water	–

Estimation of Pollution Index of Groundwater

PIG is a highly useful technique that many studies have used to assess the water quality (both surface and groundwater) [28–29]. The findings of the PIG categorization for the 22 groundwater samples obtained from the research area are shown in Table 5, and a distribution map of PIG is shown in Figure 15. The range of PIG values is 0.98 to 1.98, with an average value of 1.36. (Table 6). According to PIG's classification, 5% of the groundwater samples that were collected are in a zone with insignificant pollution, while 77% and 18% of the samples are in low and moderate contamination zones, respectively.

Table 5. Classification of Pollution index of groundwater

Range of PIG	Pollution class	No. of samples (%)
<1.0	Insignificant pollution	5%
1.0–1.5	Low pollution	77%
1.5–2.0	Moderate pollution	18%
2.0–2.5	High pollution	–
>2.5	Very high pollution	

Table 6. Pollution Index of groundwater samples.

S.N.	Class	PIG
1	Moderate pollution	1.84
2	Moderate pollution	1.71
3	Low pollution	1.23
4	Low pollution	1.15
5	Low pollution	1.42
6	Low pollution	1.17
7	Low pollution	1.22
8	Low pollution	1.07
9	Low pollution	1.42
10	Insignificant pollution	0.98

S.N.	Class	PIG
11	Low pollution	1.24
12	Moderate pollution	1.52
13	Moderate pollution	1.98
14	Low pollution	1.34
15	Low pollution	1.13
16	Low pollution	1.22
17	Low pollution	1.4
18	Low pollution	1.44
19	Low pollution	1.4
20	Low pollution	1.38
21	Low pollution	1.36
22	Low pollution	1.37

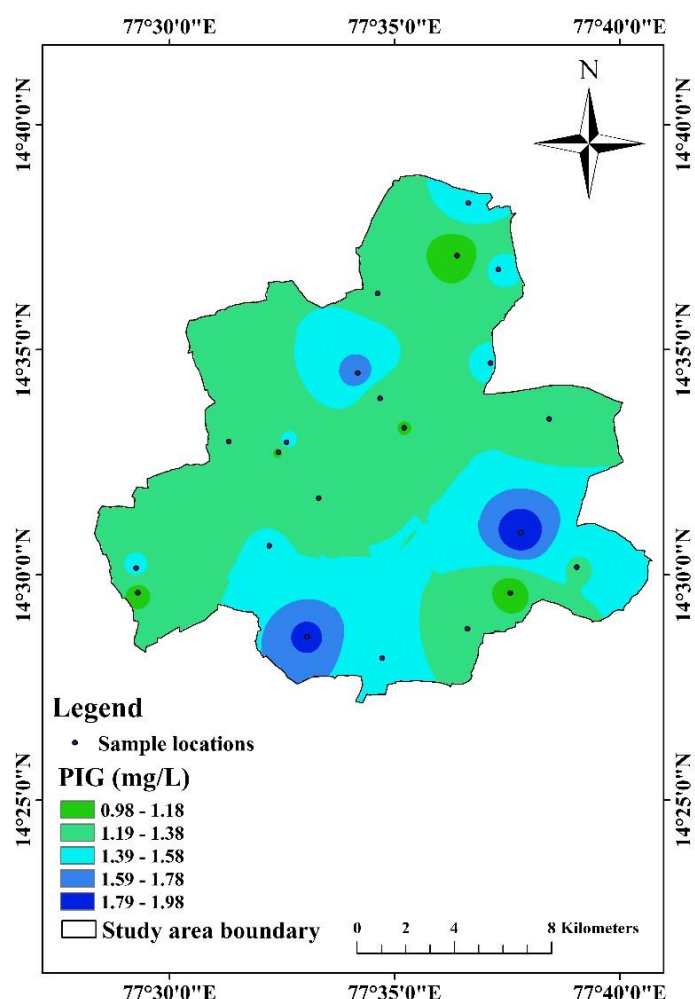


Figure 15. Spatial distribution map of PIG

Gibbs diagram

According to Gibbs (1970), the anions (Cl^-)/($\text{Cl}^- + \text{HCO}_3^-$) and cations ($\text{Na}^+ + \text{K}^+$)/($\text{Na}^+ + \text{K}^+ + \text{Ca}^{2+}$) of groundwater samples were plotted individually against TDS (Figure 16) [30, 31, 32]. The Gibbs diagram depicts three different types of domains that regulate the chemistry of groundwater. These include evaporation, precipitation, and rock dominance. All groundwater samples in the research area are found in a rock-dominated area.

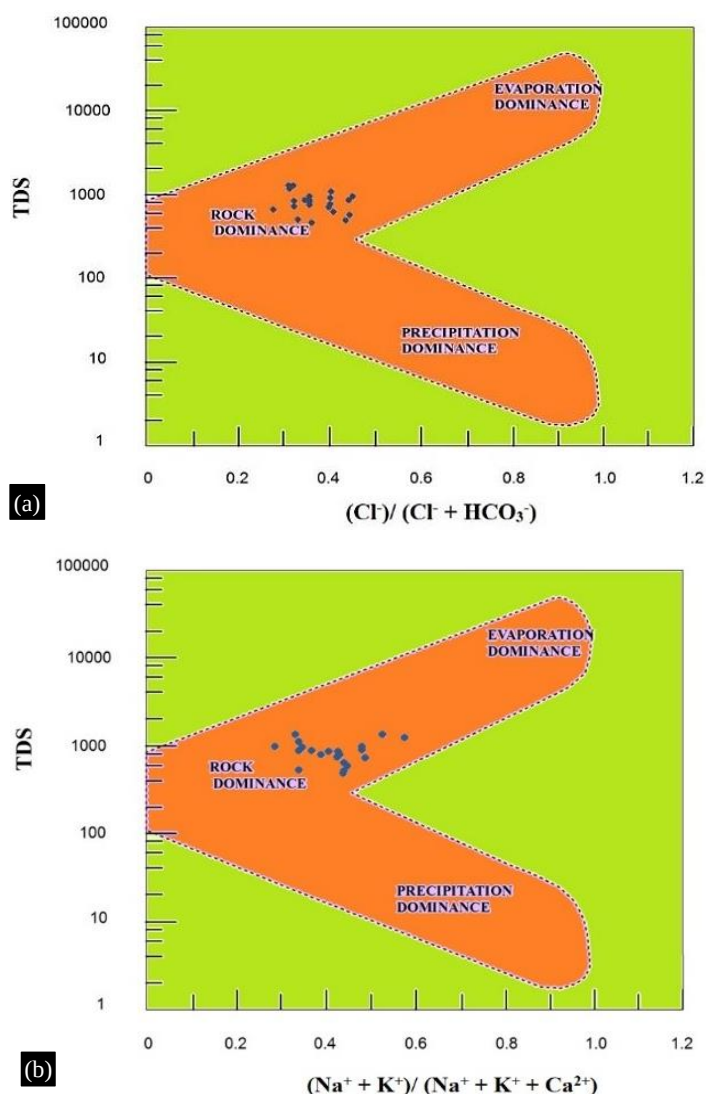


Figure 16. Gibbs diagram of Major anions and cations Vs TDS

CONCLUSION

The main objective of this work was to integrate WQI, PIG, and GIS to assess the quality of groundwater for drinking. Groundwater is reported to range from being slightly alkaline to slightly acidic. The outcomes of the chemical analyses were contrasted with those required by the WHO and BIS. In the studied region, the groundwater geochemistry shows most dominant anions are $HCO_3^- > SO_4^{2-} > Cl^- > NO_3^-$, most dominant cations are $Na^+ > Ca^{2+} > Mg^{2+} > K^+$. Based on PIG classifications, Insignificant contamination was detected in 5% of groundwater samples (class 1), low pollution was detected in 77% of groundwater samples (classes 1–1.5), and moderate pollution was detected in 18% of groundwater samples (1.5–2.0). Utilizing WQI In the research area, 55% of groundwater samples are appropriate for potable, whereas 45% are not. In the Gibbs plot, the field of rock dominance encompasses all groundwater samples. According to the spatial variation of groundwater quality analysis carried out using the GIS technique, the bulk of the groundwater samples in this area slightly meet the standards for drinking water. The current condition of groundwater indicates the need for regular checks and efficient management techniques.

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